LAB 1 – WEARABLE FOR RESPIRATION RATE

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Overview

- **Introduction**
- **Data Acquisition**
- **Wearable**

Introduction

Goal: develop a wearable that monitors respiration rate

Method: monitor chest movements during inhalation and exhalation

Deliverable:

- A short report per group that includes plots produced throughout the lab and answers to questions in different tests. **All sections that require a part added to the report are written in red and bold.**
- To be submitted on Moodle.
- Deadline 28.10.2026, 23:59.

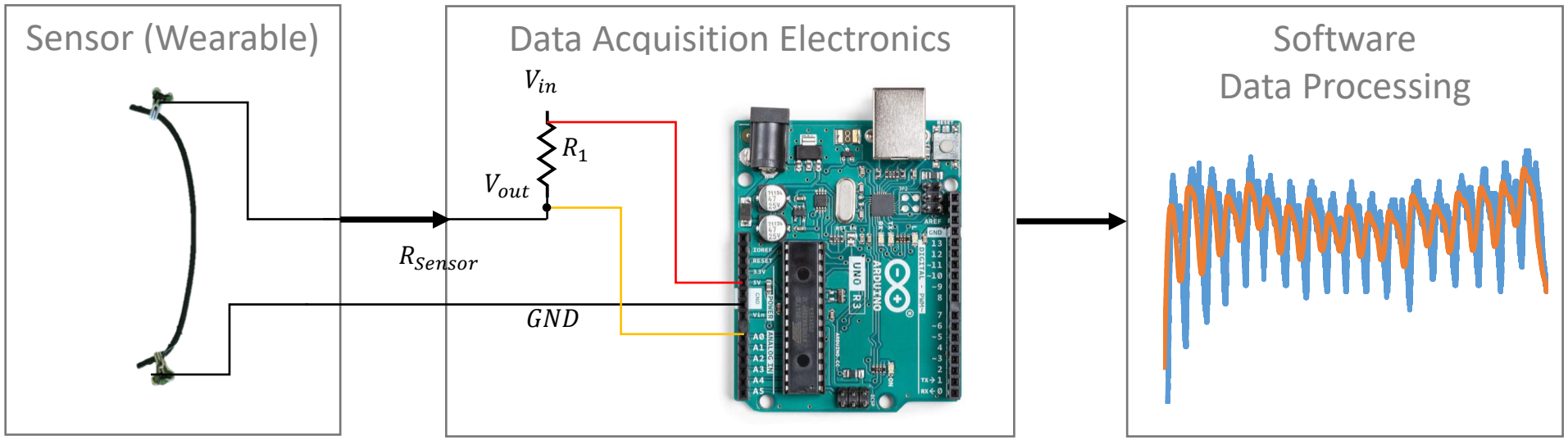
Introduction

Requirements:

Laptop with a USB port with the following software installed:

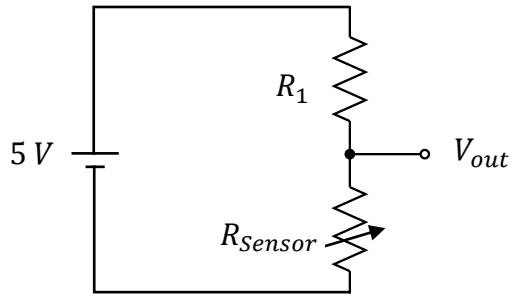
1. CoolTerm (can be downloaded here: https://freeware.the-meiers.org/)
2. Arduino IDE (can be download here: https://www.arduino.cc/en/software)
3. Python or another data processing tool

Wearable Prototype

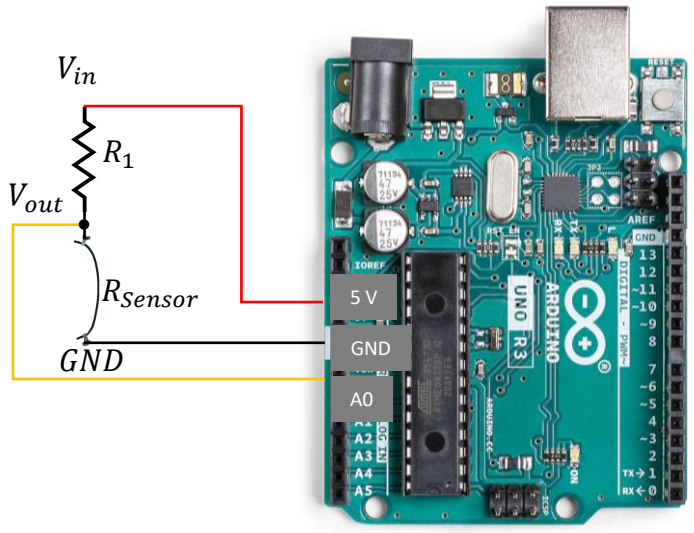


Sensor Data Acquisition

Voltage divider circuit: $$V_{out} = \frac{R_{sensor}}{R_1 + R_{sensor}} V_{in}$$

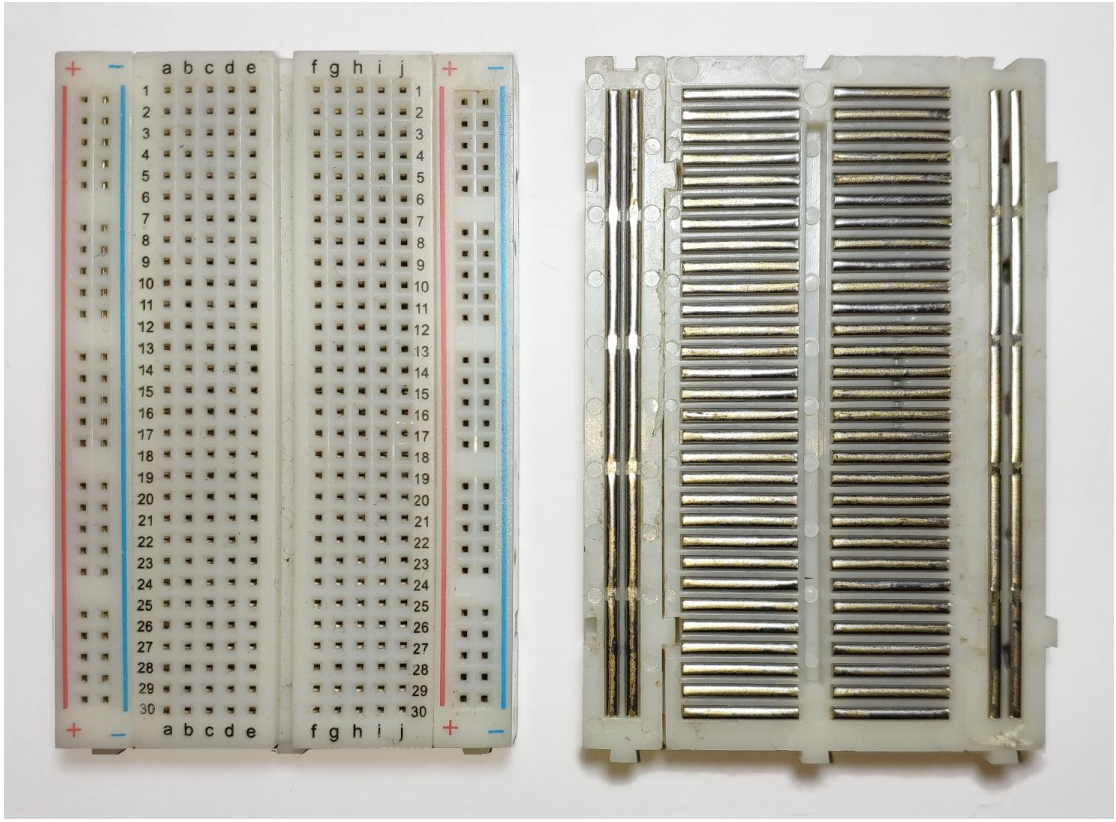


Microcontroller (Arduino) to read signal (voltage) and send it to the computer



How to use a Breadboard

- Solderless, flexible for prototyping
- Conductive metal strips connect anything inserted in that row
- Power source connect to the power rails



Top and bottom view of a breadboard, showing the connections between the holes

Source: Guhuru https://commons.wikimedia.org/w/index.php?curid=97240411

Common issues:

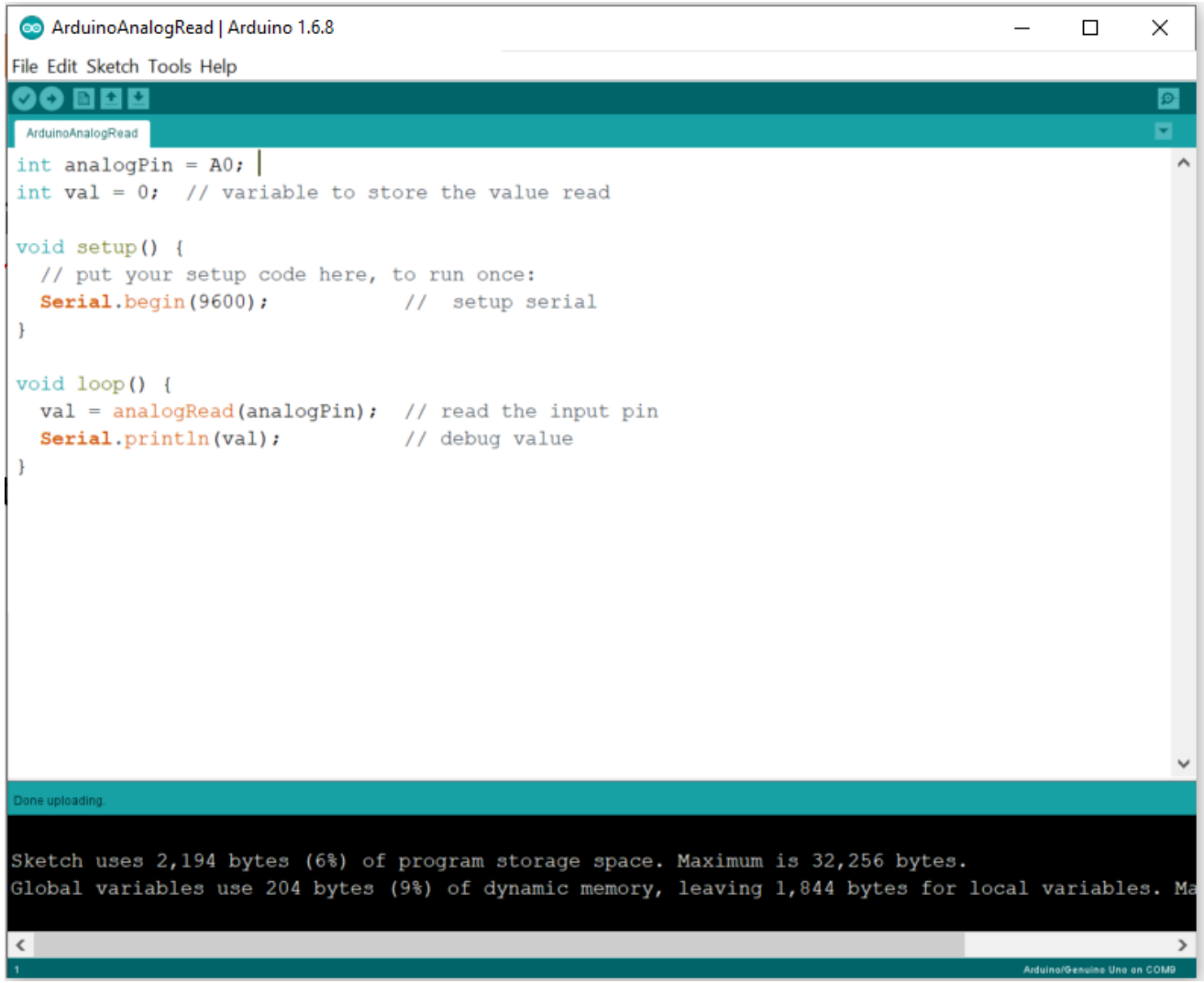
- Leads/wires are not firmly pushed into the holes
- Row mismatch of the leads/wire

Arduino Program

To read values at the V_{out} pin

1. Open the Arduino IDE
2. Assign the V_{out} pin (e.g., A0) to an integer called 'analogpin': *int analogPin = A0;*
3. Define an integer (e.g., called 'val') to keep the values read from the V_{out} pin.
   - Initialize the integer to 0. *int val = 0;*
4. Begin serial port and set the data rate
   - In setup(), add *Serial.begin(9600);* to set the data rate (baud rate) to 9600 (bits/second).
5. Read and print V_{out} pin and save the value in the integer you defined in step 3.
   - In loop() add:
     - *val = analogRead(analogPin); // read the input pin*
     - *Serial.println(val); // print value*

Arduino Program



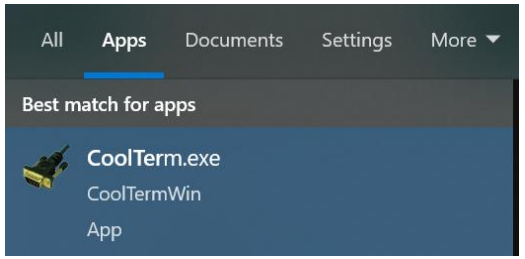
Arduino Program

6. Verify the program (sketch → verify)
7. Upload the program to the Arduino
   - Tools → Board: make sure the correct type of Arduino board is selected (Uno in our case)
   - Tools → Port: make sure the correct port is selected (to check, in device manager → Ports, plug and unplug Arduino to see which port changes)
   - Upload the program (sketch → upload)
8. Check the values being read
   - Open serial monitor (Tools → Serial Monitor/Serial Plotter). Make sure the baud rate is the same as the one you used in the program.
   - Check values. You can start stretching the sensor and the values should change. Alternatively, you can connect the analog pin to GND or 3.3V or 5V pins of the Arduino and see the values changing accordingly.
9. Close the serial monitor. If the serial monitor is left open, the next steps won't work.

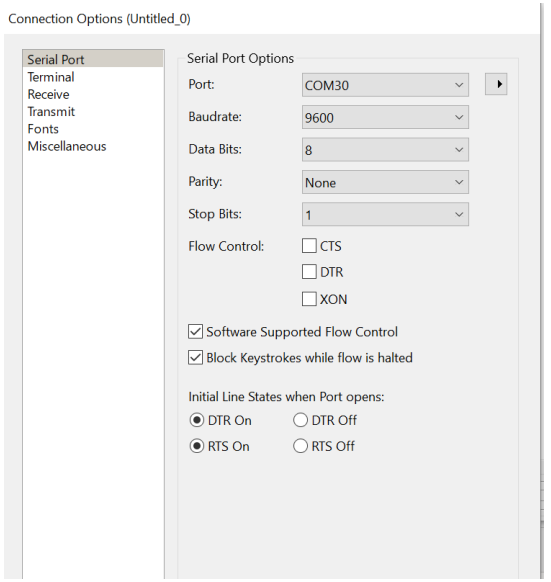
Save Signal/Coolterm

To read save signals

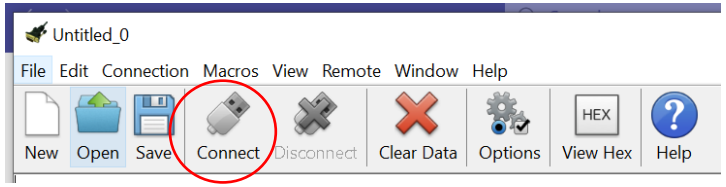
1. Make sure the serial monitor on the Arduino IDE is closed
2. Open the Coolterm application



3. Click on Connections→ Options and select the correct port and baudrate, press Enter

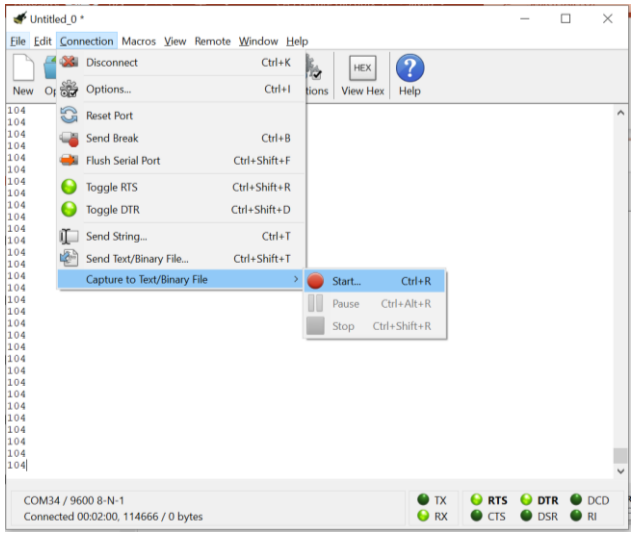


4. Click on Connect.

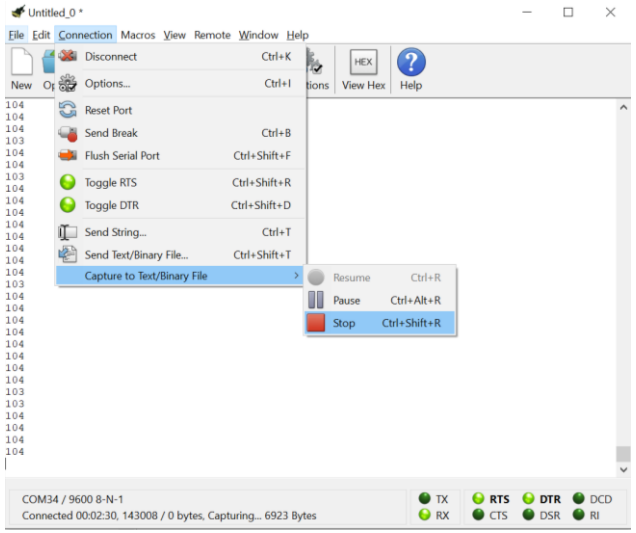


Save Signal/Coolterm

4. You should see signal values in the program now
5. To save data, go to Connections → Capture to Text/Binary File → Start and select the location of the file to be saved



6. When your recording is done, go to Connections → Capture to Text/Binary File → Stop
7. Click on disconnect to disconnect from the Arduino



Sensor Tests

At this point you should be able to read sensor values (through a voltage divider circuit) and save them to file.

In this step, you will test sensor outputs with respect to strain:

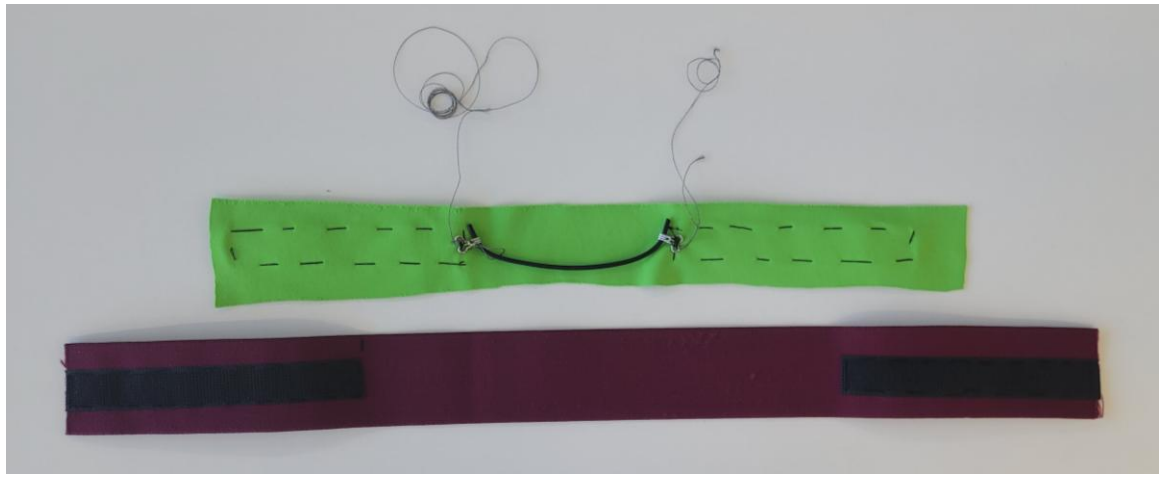
1. Using the formula for the voltage divider circuits, **find the relationship between the output read by the Arduino and the resistance of the sensor.** To find the relationship between Arduino ADC output and physical voltage, connect the analog input pin to the max voltage range: at GND(0 V) and V_{in} $(5V)$and note the correspondence to the ADC values.
2. Use a ruler to strain the sensor (10 cm segment) to 20% strain slowly and bring it back to its original state. Repeat this test 3 times, save data to file, and **plot sensor output (resistance in ohms using the result of step 1) vs. sample number. Is the sensor behavior as expected?**

Sensor Tests

3. Condition the sensor by straining it to about 50% strain 10-20 times. Now repeat the previous test. **Plot the data (sensor resistance vs. sample number). Did sensor behavior change? Provide a brief explanation of your observation.**
4. Start measuring at 0% strain, then slowly strain the sensor to 10%, then to 20%, and to 30% strains. After 30%, bring the sensor back to 20%, then 10%, and finally to 0% strain. **Keep the sensor at each strain level for 5 seconds**. Repeat 3 times. **Plot sensor response. What is the average and standard deviation of sensor output (resistance) for each strain level (what is it in extension, what is it in recovery)? Plot the calibration curve for the extension part (see graph in appendix).**

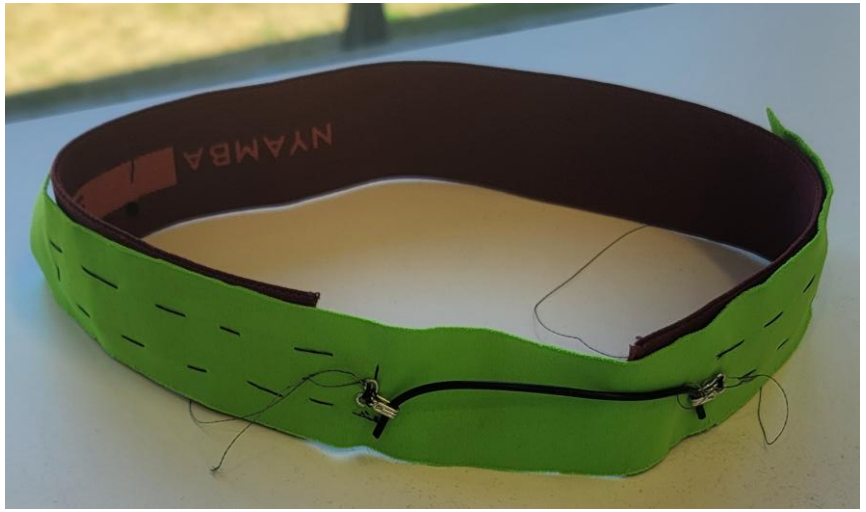
Sensor Integration into a wearable device

1. Sew clasps to the elastic fabric about 10 cm apart.
2. Attach sensor to the clasps so that the sensor is a bit loose like in the picture. Secure the connections with pliers.
3. Attach (tie tightly) conductive threads to clasps for connection.
4. Attach Velcro to the elastic fabric on the two sides of the sensor



Sensor Integration into a wearable device

5. Cut the exercise belt to size. Attach Velcro to the appropriate place so that when added to the elastic fabric (like the image below), the belt is tight around the chest.
6. Connect the two conductive threads using alligator cables and jumper wires to the voltage divider circuit



Wearable Data Collection

At this point you should have a wearable that uses a strain sensor to monitor movements of the chest and consequently respiration.

In this step you will collect respiration data using the wearable:

1. Wear the belt on your chest. Test signals when worn on clothing, under clothing, and with various levels of tightness. Test if signals can monitor breathing with various depths of breath taken. **What are your observations?**
2. Take deep breaths for 1-2 minutes while collecting respiration data. Count how many breaths you have taken. Repeat with less deep breaths. **Plot raw respiration signals for both types of breathing (sensor resistance vs. sample number).**
3. **Filter data from previous step (e.g., moving average filter) and find the number of peaks. Does this match the number of breaths you counted?**

Appendix

	0% extension	10% extension	20% extension	30% extension	20% recovery	10% recovery	0% recovery
Average	1146.55	1180.54	1230.76	1260.04	1270.12	1240.52	1235.02
Std	14.08	26.75	21.21	14.41	27.61	30.65	28.43

Strain	Average resistance	Std dev
0	1146.55	14.08
0.1	1180.54	26.75
0.2	1230.76	21.21
0.3	1260.04	14.41

